Analyzing Smart City Bike Sharing Data using Power BI

Data Transformation, Modeling & Visualization

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1. Project Overview:

The project based on public bike-sharing systems; this system generates continuous data from hundreds of stations across different cities. The task is to analyze this real-time bike station dataset to understand station performance, usage efficiency, and operational patterns across cities.

Objectives: Applying Power BI Skills from data cleaning and modeling to DAX and dashboarding to turn raw data into actionable urban mobility insights**.**

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2. Dataset Description:

This real time bike station dataset from hundreds of stations across the different cities.

Column Name	Description
Number	Bike Number
Name	Name of the station
Address	Address of the station
Position	Position of the bike station like longitudes, latitudes
Banking	Station accept payment by banking cards.
Bonus	Gives a bonus credit for specific like return bike in same station
Contract Name	Name of the contract
Bike stands	Showing total capacity
Available bike stand	The number of empty docks currently free for riders to return and lock a bike.
Available Bikes	The count of functional bikes currently docked and ready to be rented right now.
Last Update	Showing last updated date and time.

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3. Data Sources:

- **Source Description and Timeline:** Real time bike station data and 2022-2025
- **Domain:** Smart Transportation and Mobility.

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4. Problem Statement:

1. Descriptive Analytics (What is the current situation?)

- How many total bikes and empty parking spots do we have across all cities right now?
- How many stations allow riders to pay with a credit card directly at the machine?

2. Diagnostic Analytics (Why is this happening?)

- Why are some stations always completely empty while others are totally full? Does giving riders "free minutes" (bonus stations) actually help keep stations balanced?
- Why are certain cities running out of bikes faster than others? Is it because they don't have enough parking docks, or because stations are broken down and closed?

3. Predictive Analytics (What will likely happen next?)

- Looking at the time of day, when are stations in busy downtown areas most likely to run completely out of bikes?
- If a city grows and adds new neighborhoods, which areas will face the biggest shortage of bike parking spots next month?

4. Prescriptive Analytics (What should we do about it?)

- Where should the delivery trucks go first today? Which 10 stations urgently need more bikes dropped off, and which ones need bikes taken away?
- Where should we add more "bonus stations" to trick or motivate riders into moving the bikes for us, saving us money on delivery trucks?

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5. Tools & Technologies:

- **Power BI:** Data Cleaning, Transformation, Data Modeling, Dax Calculations, Visualization, and Interactive dashboard creation.

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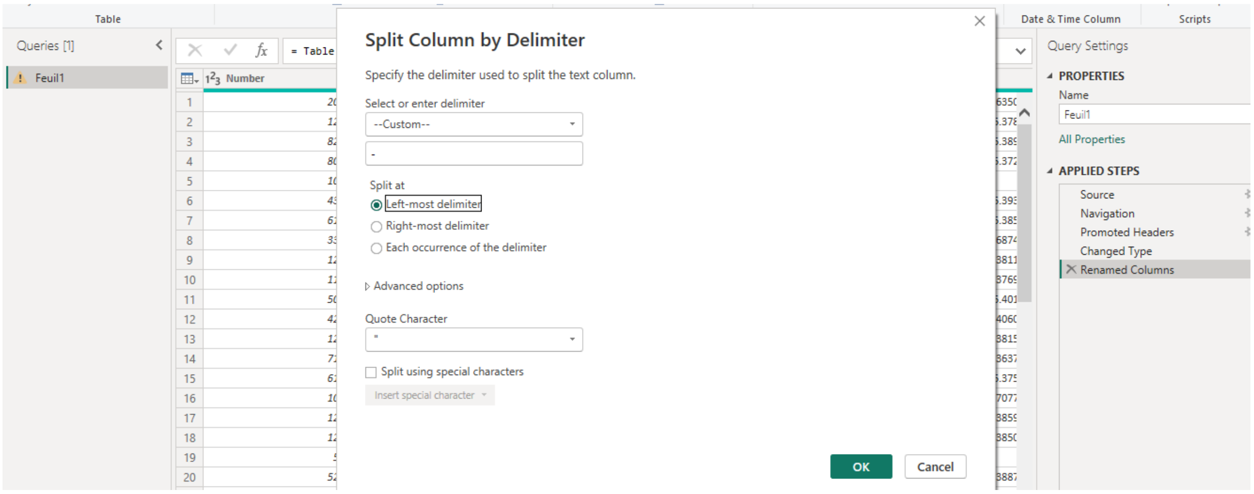
6. Data Pre-processing:

Task Performed:

- **Data Cleaning & Transformation:** Handling missing values, Remove duplicates, Standardized format,

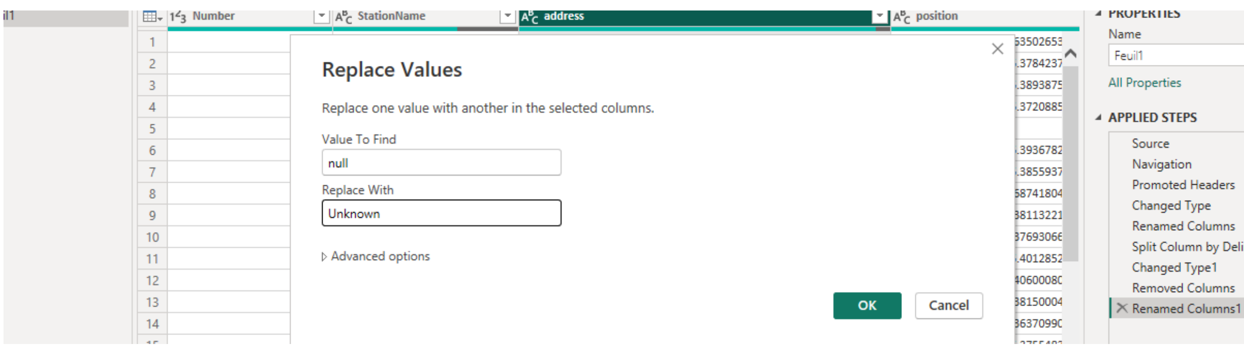
Standardized Format:

Split the address which has station number by using delimiter and split the Position column as Longitudes, Latitudes.



Handling Missing values:

Handling null values of Address column as "Unknown".



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7. Data Modelling and DAX:

DAX Calculation:

- **Total capacity:**
  - Total Capacity = SUM (Data Table [Bike Stands])
- **Total station count:**
  - Station Count = COUNT (Data Table [Station Name])
- **Station fill Rate:**
  - Station Fill Rate = DIVIDE (SUM ('Data Table'[Available Bike Stands]), SUM ('Data Table' [Bike Stands]), 0)
- **Closed Station:**

- Closed Station = CALCULATE (SUM (Data_Table [Bike Stands]), Data_Table [Station status] = "CLOSED")
- **Active Empty station:**
  - Open empty station = CALCULATE (SUM (Feuil1[Available Bike Stands]), Feuil1[Station status] = "OPEN")
- **Parking Shortage Risk:**
  - Parking shortage Risk = DIVIDE (SUM (Data_Table [Available Bike Stands]), SUM (Data_Table [Bike Stands]), 0)
- **Truck Delivery Risk =**
  - Truck Delivery Risk = ABS(SUM('Data_Table'[Available Bikes]) - SUM('Data_Table'[Available Bike Stands]))

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8. Analysis and Visualizations (Power BI):

Visualizations:

Dashboard 1:

- **KPI Card:** Total Capacity of station, Available Bike stands, Available Bikes.
- **Line Chart:** Analyzing available bike stands by hours.
- **Stacked Bar Chart:** Analyzing running out of bikes by contract Name
- **Scatter Plot:** Analyzing some stations always completely empty while others are totally full? bonus stations actually help keep stations balanced
- **Slicer:** To view by station name.

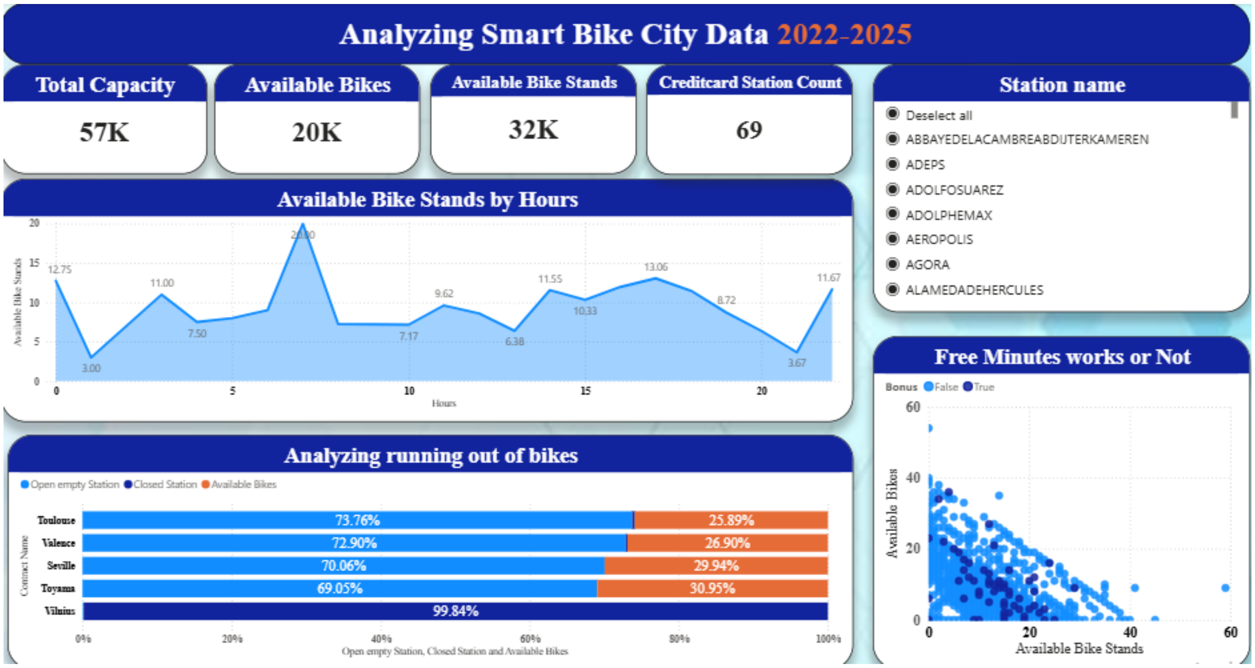
Dashboard 2:

- **Cluster Bar Chart:** Analyzing parking shortage risk by station name.
- **Slicer:** To view by station name**.**

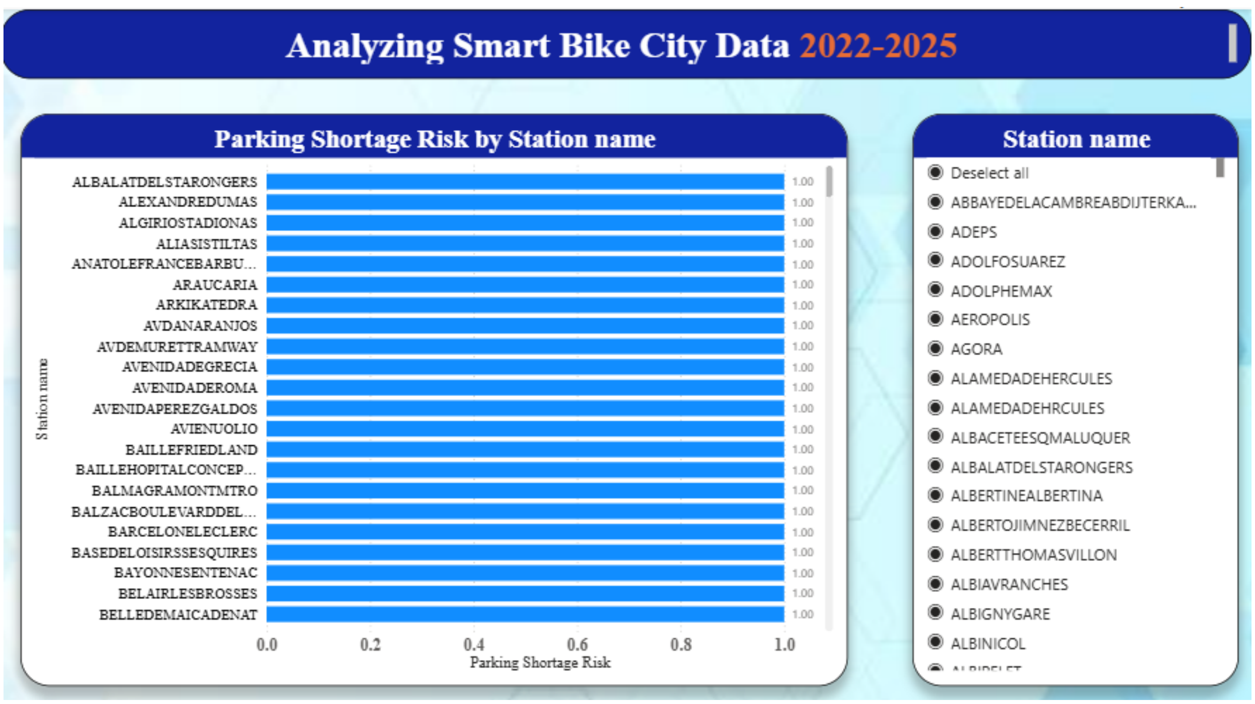
Dashboard 3:

- **Table chart:** Top 10 bottom of Minimum available bikes by station name
- **Table chart:** Parking shot risk by station name
- **Matrix chart:** Truck delivery risk by station name
- **Slicer:** To view by station name.

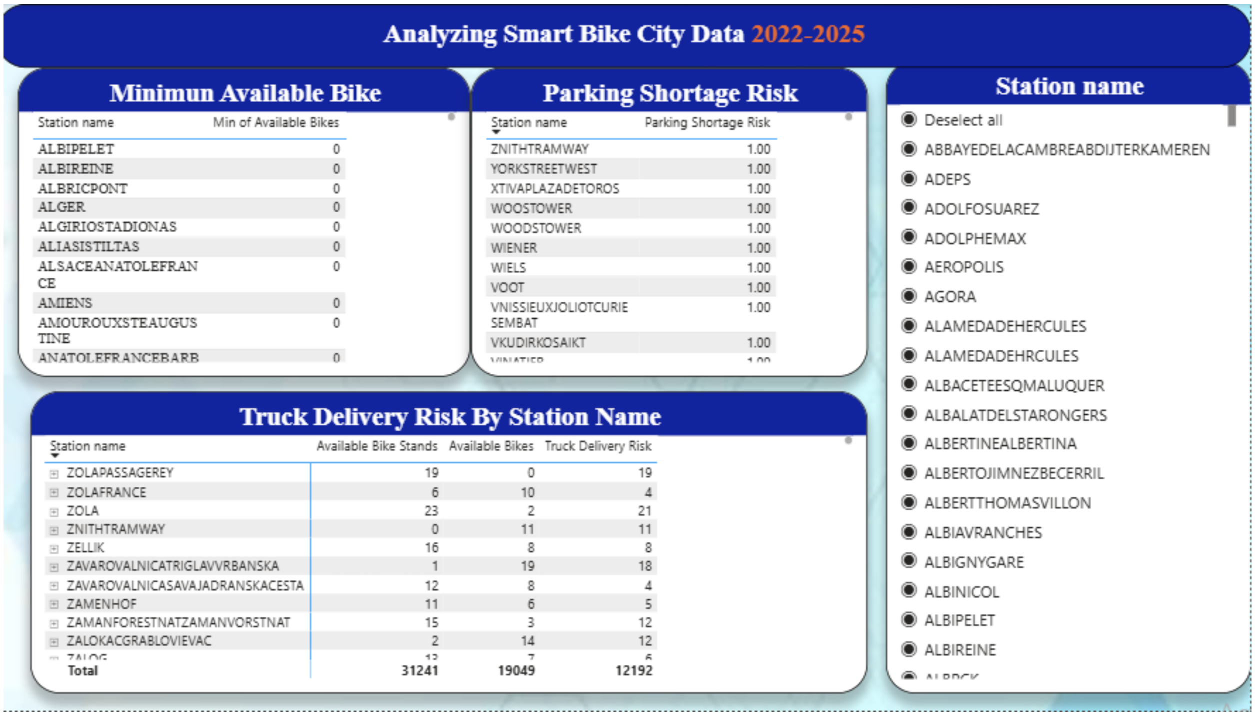
Dashboard 1:



Dashboard 2:



Dashboard 3:



9.Insights & Conclusion:

- Calculated Total capacity, Total available bikes and total available bikes stands.
- Findings the how many riders are paid with credit card directly at the machine.
- **The "Free Minutes" Fallacy (Incentive):** No, it is not working effectively,
- Diagnostic analytics proved that the "Bonus Station" (free minutes) incentive is currently failing to maintain network equilibrium. Scatter plot mapping reveals that both `Bonus = True` and `Bonus = False` stations stay heavily polarized at extreme boundaries choked completely full or drained entirely empty proving that the existing bonus structure does not alter rider behavior enough to organically balance the system.
- **Dual-Core Drivers of Bike Shortages:** The root cause behind cities running out of bikes differs entirely by region. Most contracts (Toulouse, Valence, Seville, Toyama) suffer strictly from extreme demand imbalances, leaving stations fully functional but drained of bicycles (~70% open empty docks). Conversely, Vilnius represents a pure infrastructure and maintenance crisis, running out of bikes because a massive 99.84% of its network is stuck in a closed or deactivated station status.
- **Predictive Rush Hour Timelines:** Downtown micro-mobility hubs suffer from a highly predictable daily risk cycle. Stations face their highest depletion risk during late-night leisure periods (1:00 AM) and late-evening social spikes (9:00 PM), whereas the morning commuter rush (7:00 AM) acts as a peak supply reset window.

- **Proactive Expansion Planning:** Top-N filtered modeling isolated the specific target areas facing immediate gridlock as the city footprint grows next month. Planners can now isolate the top 24 highest-risk neighborhoods that sit at a maximum 1.00 Parking Shortage Risk score to prioritize immediate physical dock expansions.
- **Smart Truck Routing:** Dispatch trucks on a strict high-priority loop to drop bikes off at the 10 completely empty stations (0 bikes) and remove inventory from the 10 overflowing hubs (1.00 risk).
- **Self-Healing Network:** Add new bonus incentives at maximum-risk hubs where the gap between bikes and empty docks is widest, motivating riders to move bikes themselves to eliminate fleet trucking costs**.**

Conclusion:

The end-to-end data analysis was executed entirely within Power BI, proving its power from raw data ingestion to advanced visual reporting. Power BI Query Editor was utilized to clean, transform, and preprocess the real-time dataset. Within the data model, calculated fields and DAX measures were built to drive descriptive, diagnostic, predictive, and prescriptive dashboards.

This centralized Power BI pipeline successfully isolated severe network polarization, exposed the failure of the current bonus system, pinpointed specific infrastructure breakdowns in Vilnius, and delivered a live routing blueprint to eliminate fleet trucking costs.

!!!!Thank you!!!!